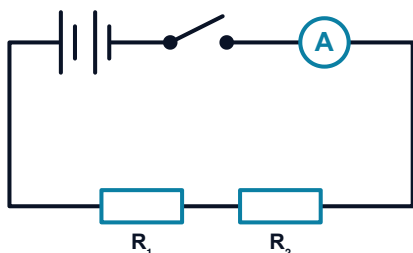


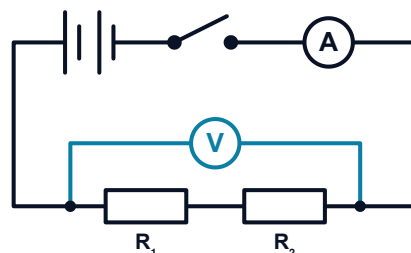
Resistors in series and in parallel

A student compares the total resistance of two resistors in series and in parallel.

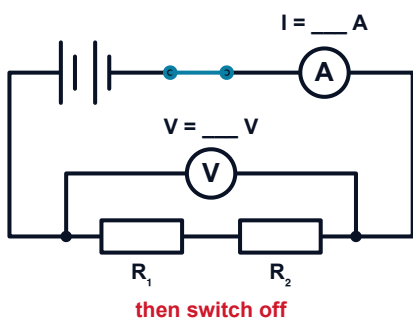
- 1** Connect the two resistors in series with a power supply and an ammeter.



- 2** Connect a voltmeter across both resistors.



- 3** Switch on, record the current and the potential difference, then switch off.



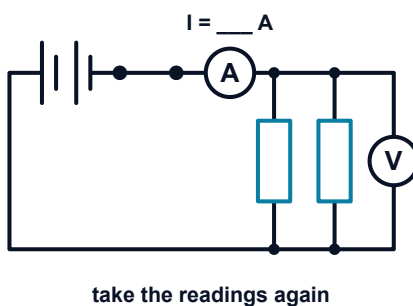
- 4** Calculate the total resistance using $R = V \div I$.

$$R = V \div I$$

potential difference (V) $\div$ current (A)

R = total resistance of the two resistors

- 5** Reconnect the two resistors in parallel and take the readings again.



- 6** Compare the two values of total resistance.

series

vs

parallel

R = ___ Ω

R = ___ Ω

which is bigger?